

MUTUAL FUND ANALYTICS PLATFORM

Capstone Project Report

Submitted By

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Role

Data Analytics Intern

Technologies Used

Python, Pandas, NumPy, SQLite, Power BI, Git, GitHub

Executive Summary

The Mutual Fund Analytics Platform is an end-to-end data analytics project developed to analyze mutual fund performance, investor behavior, and portfolio characteristics. The project integrates multiple datasets including fund master data, NAV history, investor transactions, and portfolio holdings into a centralized analytics solution.

The primary objective of the project was to build a structured ETL pipeline capable of extracting, transforming, and loading mutual fund data into a SQLite database for efficient analysis. Using Python, various data cleaning, validation, and transformation processes were implemented to improve data quality and consistency.

Exploratory Data Analysis (EDA) was performed to identify trends in SIP inflows, category-wise fund participation, investor activity, and portfolio allocation. Performance analysis was conducted using benchmark comparisons and several risk-adjusted metrics including CAGR, Sharpe Ratio, Alpha, Beta, Maximum Drawdown, and Tracking Error.

To support interactive exploration of the data, multiple dashboards were developed using Power BI. These dashboards provide insights into industry growth, fund performance, investor behavior, and market trends. The final solution demonstrates how data analytics can be used to support informed investment decisions and improve mutual fund analysis workflows.

Project Overview

Background

The mutual fund industry generates a significant amount of data related to fund performance, investor transactions, portfolio holdings, and market activity. However, this information is often distributed across multiple files and systems, making comprehensive analysis difficult.

Problem Statement

Mutual fund information was available across multiple datasets, making manual analysis difficult and time-consuming. Important information such as NAV history, investor transactions, portfolio holdings, and performance metrics existed in separate datasets.

The absence of a centralized analytical framework created challenges in evaluating fund performance, identifying investment trends, and generating business insights.

Project Objectives

The objectives of this project were:

- Build a structured ETL pipeline for mutual fund datasets.
- Clean, validate, and standardize raw data.
- Store processed data in a centralized SQLite database.
- Perform exploratory data analysis to identify important trends.
- Evaluate mutual fund performance using financial metrics.
- Analyze risk-adjusted returns and benchmark performance.
- Generate investor and portfolio insights.
- Build interactive dashboards using Power BI.
- Create a complete analytics workflow from raw data to business insights.

Project Scope

The scope of this project includes data ingestion, data cleaning, database creation, exploratory analysis, performance evaluation, risk assessment, and dashboard development. The project focuses on mutual fund schemes, NAV history, investor transaction records, and portfolio holdings data. The analysis is limited to the datasets provided during the capstone project and does not include live market integration.

Data Sources

The Mutual Fund Analytics Platform was developed using multiple datasets that capture different aspects of the mutual fund ecosystem. These datasets were integrated to create a unified analytical framework for performance evaluation, investor analysis, and portfolio assessment.

Fund Master Data

The Fund Master dataset serves as the primary reference table and contains essential information about mutual fund schemes. The dataset includes details such as scheme name, fund category, AMC name, expense ratio, risk classification, and scheme identifiers.

This dataset was used to classify funds, perform category-level analysis, and enrich analytical outputs generated during the project.

Key Attributes

- Scheme Name
- Fund Category
- AMC Name
- Expense Ratio
- Risk Level
- AMFI Code

NAV History Dataset

The NAV (Net Asset Value) History dataset contains historical NAV records for mutual fund schemes across multiple periods.

This dataset was the primary source for return calculations and performance analysis. Historical NAV values were used to evaluate growth trends, benchmark comparisons, CAGR calculations, and risk-adjusted performance metrics.

Investor Transactions Dataset

The Investor Transactions dataset contains transaction-level information representing investor activity across mutual fund schemes.

The dataset includes SIP investments, lump-sum contributions, and redemption transactions. This information was used to understand investor participation patterns and investment behavior.

Portfolio Holdings Dataset

The Portfolio Holdings dataset contains detailed information about the securities held by mutual fund schemes.

The dataset includes stock names, sectors, portfolio weights, and market values. Portfolio information was used to analyze sector allocation and portfolio concentration.

Dataset Summary

The final analytical environment consisted of multiple datasets covering mutual fund schemes, historical NAV records, investor transactions, and portfolio holdings.

The integrated dataset contained:

- 10 Source Datasets
- 40 Mutual Fund Schemes
- 46,000+ NAV Records
- 35,000+ Investor Transactions

ETL Pipeline Design

An ETL (Extract, Transform, Load) pipeline was developed to automate the process of data ingestion, cleaning, transformation, and storage.

The ETL workflow ensured that raw datasets were converted into structured analytical tables suitable for reporting and dashboard generation.

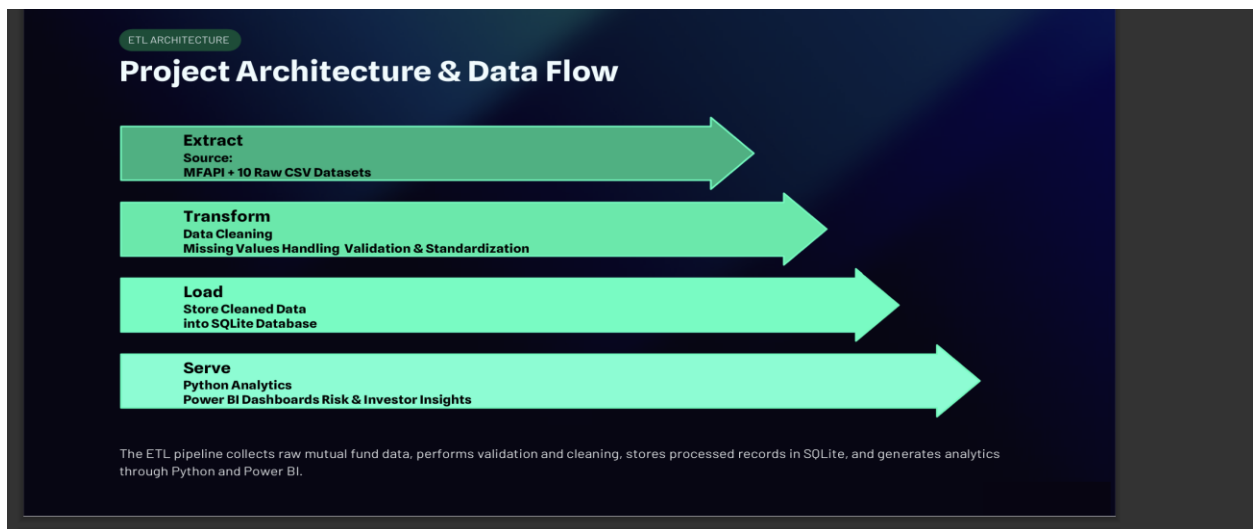


Figure 1: ETL Architecture of the Mutual Fund Analytics Platform

Extract Phase

The extraction phase involved collecting data from multiple CSV files and MFAPI sources. Raw datasets included:

- Fund Master Data
- NAV History Data
- Investor Transactions Data
- Portfolio Holdings Data

The extracted datasets were stored in a staging layer for further processing.

Transform Phase

The transformation phase focused on improving data quality and preparing the datasets for analysis.

Several preprocessing activities were performed:

Missing Value Handling

Data Standardization

Data Validation

Duplicate Removal

Feature Engineering

Load Phase

After transformation, the cleaned datasets were loaded into a SQLite database.

SQLite served as the centralized storage layer for the project and enabled efficient querying of mutual fund data.

Benefits of using SQLite included:

- Lightweight Database Architecture
- Faster Query Execution
- Easy Integration with Python
- Efficient Dashboard Connectivity

The database became the primary source for all analytical calculations and Power BI dashboards.

Data Cleaning and Validation

To ensure data quality and consistency, all datasets were cleaned and validated before analysis. Missing values, duplicate records, and data inconsistencies were identified and corrected during preprocessing.

Duplicate Record Removal

Duplicate records were identified and removed to improve data accuracy and reporting reliability.

Validation Checks

- AMFI code validation
- Category mapping verification
- Transaction consistency checks
- Portfolio holding validation.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand the structure of the data and identify important trends, patterns, and relationships.

EDA helped uncover valuable insights related to investor behavior, fund performance, category participation, and industry growth.

The analysis focused on understanding market activity, investment patterns, and category-level performance.

SIP Growth Trend Analysis

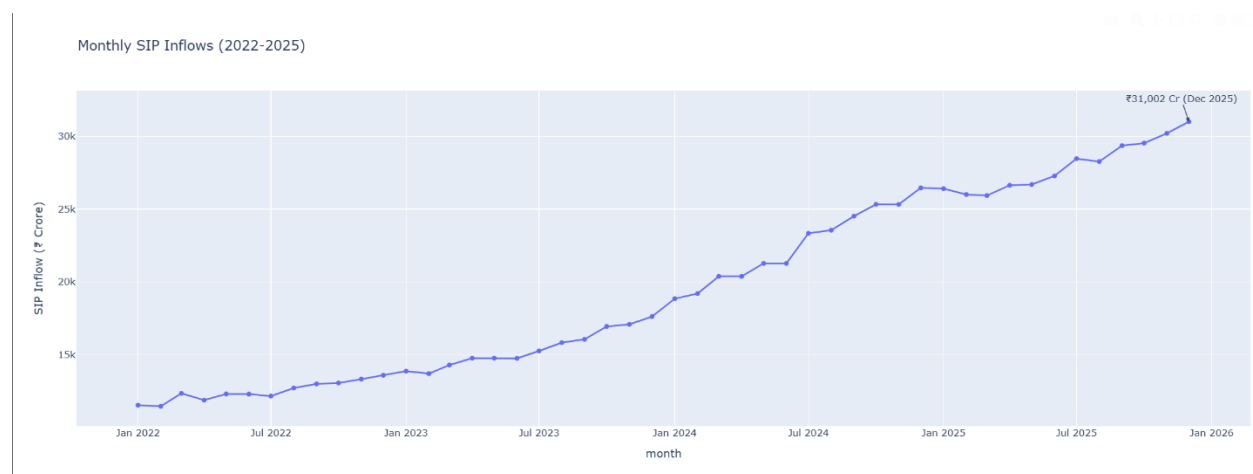


Figure 2: Monthly SIP Inflow Trend

Systematic Investment Plans (SIPs) have become one of the most popular investment methods among retail investors. Therefore, analyzing SIP trends was an important component of this project.

Monthly SIP contributions increased from approximately ₹12,000 Crore to ₹31,002 Crore, indicating substantial growth in investor participation.

Key Observations

- SIP inflows increased consistently during the analysis period.
- Retail investor participation showed strong growth.
- December 2025 recorded the highest SIP inflow.
- Long-term investment behavior continued to strengthen.

Category-Wise Inflow Analysis

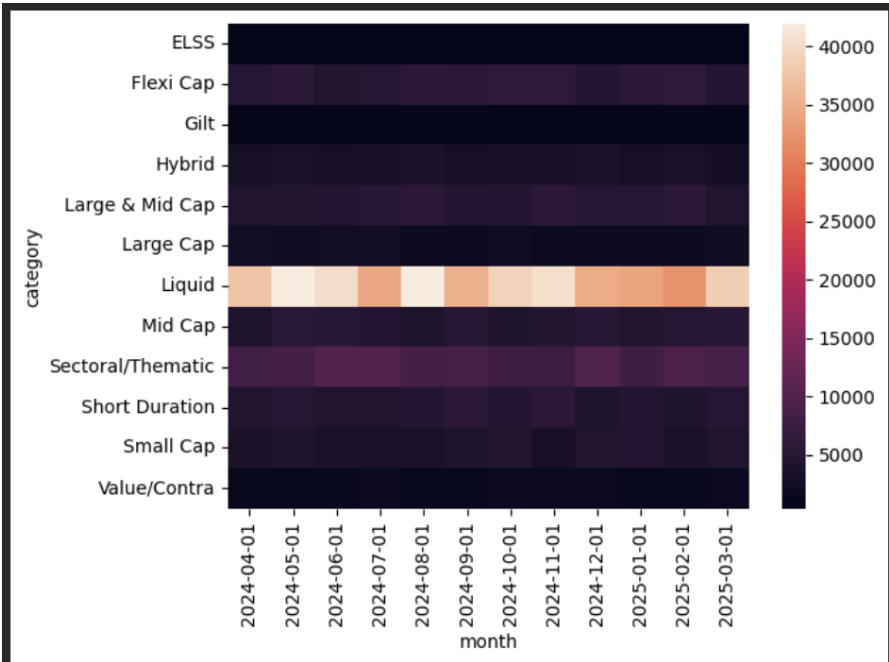


Figure 3: Category-wise Inflow Analysis Heatmap

Finding:

Liquid and Equity-oriented funds attracted the highest inflows, while debt categories remained relatively stable.

Key Observations

- Equity funds attracted the largest share of investments.
- Debt funds remained comparatively stable.
- Hybrid funds exhibited seasonal fluctuations.
- Investor preferences shifted towards equity-oriented schemes during favorable market conditions.

Portfolio Analysis and Sector Allocation

Portfolio holdings data was analyzed to understand the composition of mutual fund investments.

Sector-level allocation analysis provided insights into investment concentration and diversification patterns across schemes.

The analysis revealed that a significant portion of investments was concentrated within a limited number of sectors, highlighting the importance of diversification when evaluating mutual fund portfolios.

Portfolio concentration was further assessed using the Herfindahl-Hirschman Index (HHI), which measures the concentration of holdings within a portfolio.

Funds with higher HHI values exhibited greater concentration risk, whereas funds with lower HHI values demonstrated stronger diversification characteristics.

Key Observations

- Financial and banking-related sectors represented significant portfolio allocations.
- Portfolio concentration varied across schemes.
- Diversification levels differed substantially among funds.
- HHI analysis helped identify concentrated and diversified portfolios.

Investor Behavior Analysis

Investor transaction data was analyzed to understand participation trends and transaction behavior.

The analysis included SIP investments, lump-sum investments, and redemption activities.

Results indicated growing investor engagement and increasing participation in systematic investment plans.

The findings suggest that investors increasingly prefer disciplined and long-term investment strategies over short-term trading behavior.

Key Observations

- SIP participation continued to increase.
- Investor activity remained consistent across multiple categories.
- Long-term investment behavior was dominant.
- Transaction data revealed strong retail participation in mutual funds.

EDA Summary

The exploratory data analysis phase provided valuable insights into investor behavior, industry growth, category preferences, and portfolio composition.

The findings revealed strong growth in SIP investments, increasing mutual fund adoption, and significant participation in equity-oriented investment products.

These insights formed the foundation for the performance analysis and risk assessment conducted in subsequent phases of the project.

Performance Analytics

Performance analysis was conducted to evaluate the growth, consistency, and risk-adjusted returns of mutual fund schemes. The objective was to identify funds that delivered superior returns while maintaining acceptable risk levels.

The analysis involved benchmarking mutual fund schemes against market indices and calculating various performance metrics commonly used in the investment industry.

Benchmark Comparison Analysis

To evaluate fund performance, the top-performing mutual funds were compared against benchmark indices including NIFTY 50 and NIFTY 100.

A normalized performance approach was adopted where all funds and benchmark indices were rebased to a common starting value of 100. This allowed direct comparison of growth trajectories across multiple schemes.

The comparison revealed that several mutual funds outperformed benchmark indices over the analysis period, demonstrating superior growth and return generation.

Key Observations

Several mutual funds outperformed benchmark indices over the analysis period.

CAGR Analysis

CAGR was used to measure annualized growth and compare long-term fund performance.

Risk Analytics

Risk metrics were used to evaluate the consistency and downside risk of mutual fund schemes.

Sharpe Ratio Analysis

Sharpe Ratio was used to compare risk-adjusted returns across mutual fund schemes. Higher values indicate better return generation per unit of risk.

Alpha and Beta Analysis

Alpha

Alpha measures excess return over the benchmark. Positive alpha indicates outperformance.

Beta

Beta measures sensitivity to market movements. Values above 1 indicate higher volatility than the market

Maximum Drawdown Analysis

Maximum Drawdown measures the largest decline from peak value and helps assess downside risk.

Tracking Error Analysis

Tracking Error measures deviation from benchmark performance. Lower values indicate better benchmark tracking.

Fund Scorecard Development

A composite scorecard was created using:

- CAGR
- Sharpe Ratio
- Alpha
- Beta
- Maximum Drawdown
- Tracking Error

scheme_name	amfi_code	fund_house	category	return_3yr_pct	sharpe_ratio	sortino_ratio	alpha	beta	max_drawdown_pct
Mirae Asset Tax Saver Fund - Regular - Growth	148569	Mirae Asset MF	ELSS	13.58	0.80	1.03	0.54	0.98	-22.62
Kotak Flexicap Fund - Regular - Growth	120843	Kotak Mahindra MF	Flexi Cap	15.65	0.98	1.57	1.85	0.95	-19.50
UTI Flexi Cap Fund - Regular - Growth	102887	UTI Mutual Fund	Flexi Cap	15.34	0.96	1.37	1.79	1.00	-12.14
Nippon India Gilt Securities Fund - Regular - Growth	118636	Nippon India MF	Gilt	5.31	1.33	2.38	0.89	0.37	-2.23
SBI Magnum Gilt Fund - Regular Plan - Growth	119120	SBI Mutual Fund	Gilt	6.07	1.52	2.11	1.60	0.22	-2.30
UTI Nifty 50 Index Fund - Regular - Growth	102885	UTI Mutual Fund	Index	12.10	0.93	1.29	0.93	0.90	-24.42
Nippon India ETF Nifty 50 BeES	118635	Nippon India MF	Index/ETF	11.77	0.91	1.24	1.80	1.04	-26.75
Mirae Asset Emerging Bluechip Fund - Regular - Growth	148568	Mirae Asset MF	Large & Mid Cap	14.56	0.91	1.55	1.70	0.99	-33.15
ABSL Frontline Equity Fund - Regular - Growth	101206	Aditya Birla Sun Life MF	Large Cap	13.78	0.98	1.25	1.34	1.03	-15.07
Axis Bluechip Fund - Direct - Growth	119093	Axis Mutual Fund	Large Cap	12.14	0.87	1.54	1.43	0.87	-17.40

Power BI Dashboard Development

To improve accessibility and support interactive analysis, a series of Power BI dashboards were developed using the processed csv datasets.

The dashboards transformed analytical results into user-friendly visualizations and enabled dynamic exploration of mutual fund data.

The dashboard environment was designed to support both high-level business insights and detailed scheme-level analysis.

Dashboard Objectives

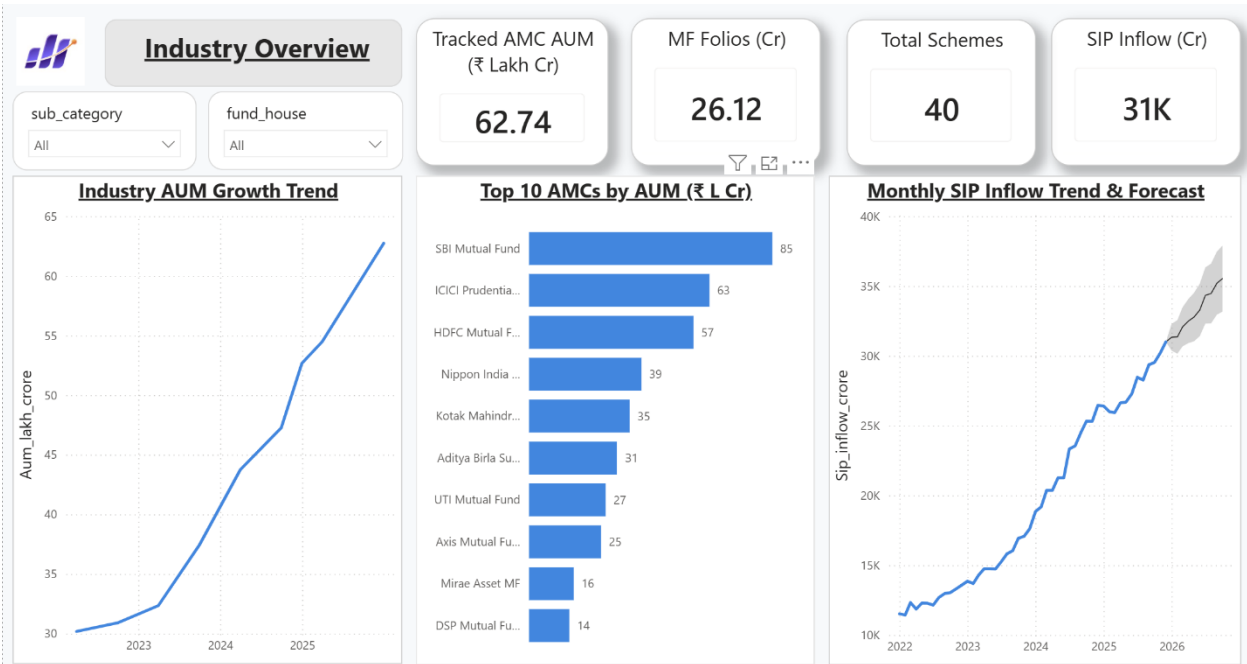
Provide interactive visualization and support data-driven investment analysis.

The dashboards served as the final presentation layer of the Mutual Fund Analytics Platform and consolidated insights generated throughout the project lifecycle.

Dashboard Overview and Visualization

Power BI dashboards were developed using processed CSV datasets and analytical outputs generated through Python. The dashboards provide interactive insights into fund performance, investor behavior, and market trends.

Industry Overview Dashboard

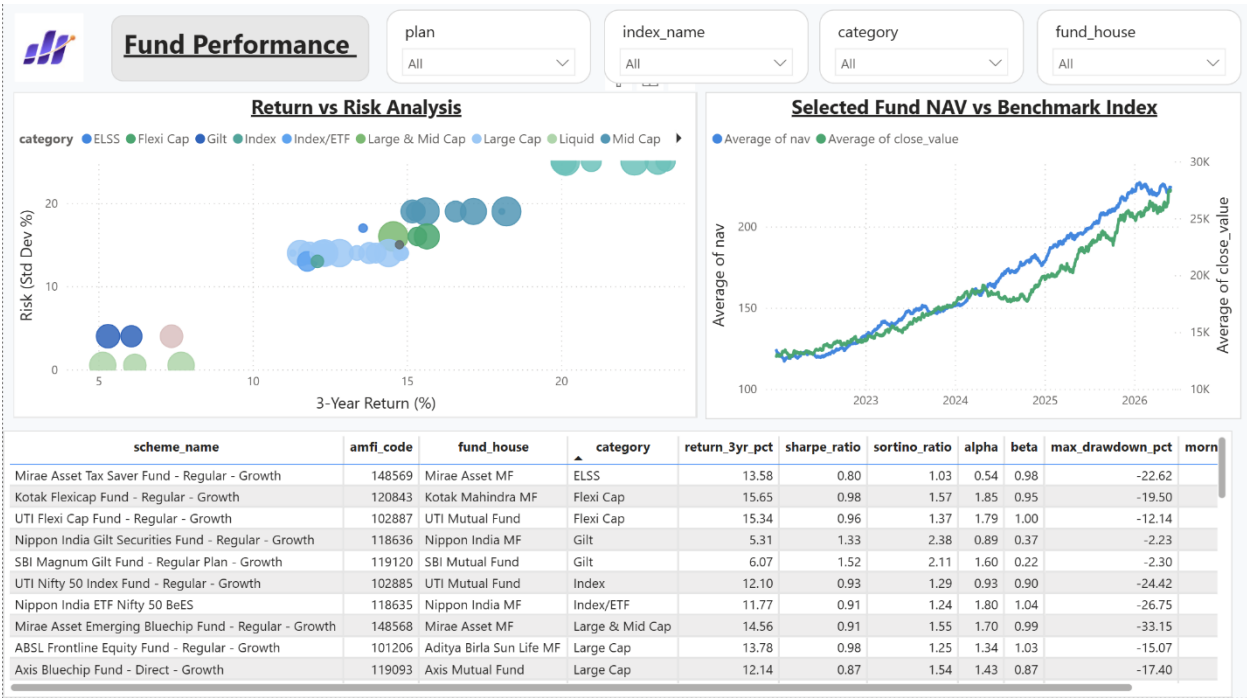


The Industry Overview Dashboard provides a high-level view of the mutual fund industry.

Key features include:

- Assets Under Management (AUM) growth trends.
- SIP inflow analysis.
- Top Asset Management Companies (AMCs).
- Category-level fund distribution.
- Industry growth indicators.

Fund Performance Dashboard



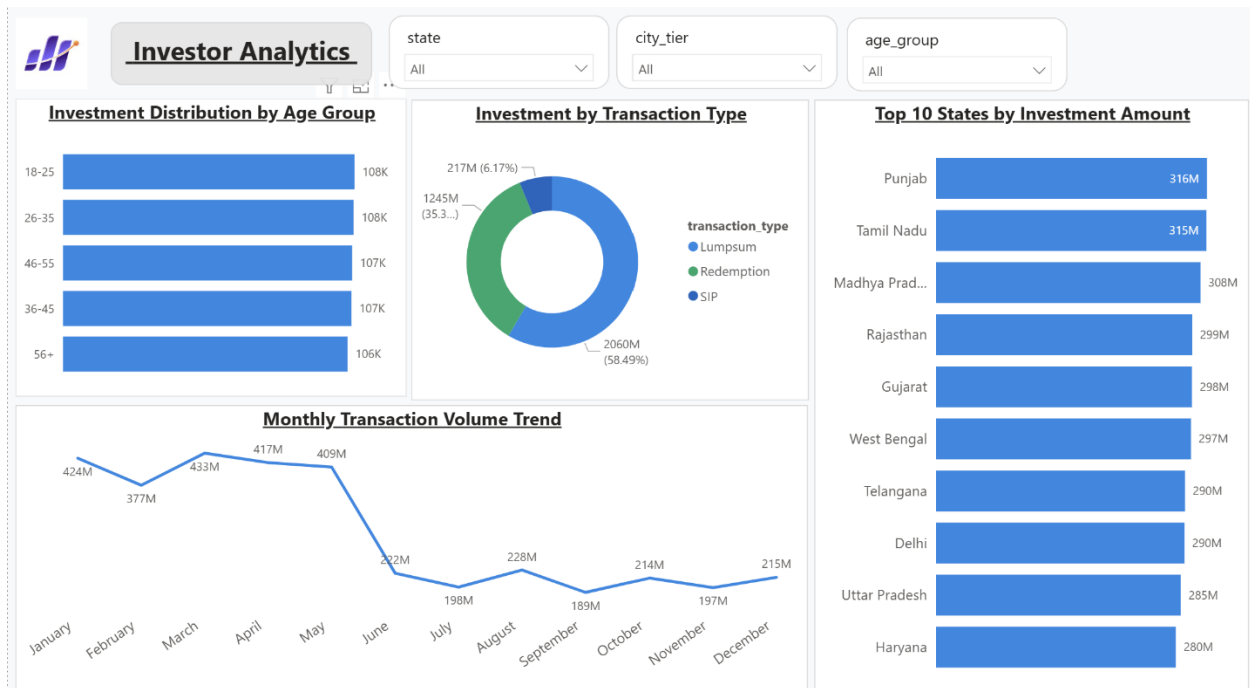
The Fund Performance Dashboard focuses on evaluating individual schemes and comparing their performance.

Key features include:

- Return versus risk analysis.
- Benchmark comparison.
- CAGR evaluation.
- Sharpe Ratio comparison.

- Alpha and Beta analysis.
- Fund ranking tables.

Investor Analytics Dashboard



Key features include:

- Age-wise investment distribution.
- State-wise investment analysis.
- SIP versus lump-sum investment patterns.
- Redemption activity tracking.
- Investor participation trends.

Key Findings

Major Findings

- Industry Assets Under Management increased significantly during the analysis period.
- SIP inflows reached record levels, indicating strong retail participation.
- Benchmark comparison revealed that multiple funds outperformed market indices.
- Investor participation continued to increase, driven primarily by SIP investments.

Challenges Faced During the Project

Data Integration Challenges

Multiple datasets required cleaning and standardization before analysis.

Data Quality Issues

Missing values and duplicate records required preprocessing and validation.

Dashboard Design Challenges

Dashboard layouts were refined through multiple iterations to improve usability.

Key Learnings

Technical Learnings

- ETL pipeline development using Python.
- Data cleaning and preprocessing techniques.
- Interactive dashboard development using Power BI.

Recommendations

- Evaluate mutual funds using risk-adjusted metrics instead of relying solely on returns.
- Focus on long-term consistency rather than short-term returns.
- Use dashboard-driven analytics to support investment decisions.

These recommendations can help investors make more informed and data-driven decisions.

Conclusion

The Mutual Fund Analytics Platform successfully integrated multiple mutual fund datasets into a centralized analytical solution. Through ETL automation, exploratory analysis, performance evaluation, risk assessment, and dashboard development, the project transformed raw financial data into meaningful business insights.

The project demonstrated the practical application of Python, SQLite, and Power BI in solving real-world analytical problems. The developed solution provides a scalable framework for mutual fund analysis and highlights the value of data analytics in financial decision-making.

Overall, the project achieved all planned objectives and delivered a comprehensive analytics platform capable of supporting performance monitoring, investor analysis, and investment research.